

Answer all questions. Calculators and mobile phones are NOT allowed.

1. [10 pts.] Let $X = (1, 1, 1, 1)$ and $Y = (1, 1, 0, 1)$ be two vectors in $\mathbb{R}^4$.
 - (a) (3 pts.) Show that X and Y are linearly independent vectors in $\mathbb{R}^4$.
 - (b) (7 pts.) Find a basis for $\mathbb{R}^4$ which contains X and Y .
2. [15 pts.] Let $P(1, 2, 3)$, $Q(0, 3, 4)$, and $R(1, 0, 2)$ be three points in $\mathbb{R}^3$.
 - (a) (8 pts.) Find the area of the triangle PQR .
 - (b) (7 pts.) Find an equation of the plane passing through the points P , Q and R .
3. [15 pts.] Let $X = (1, -2, 4)$, $Y = (3, 4, 1)$ and $Z = (-1, 1, 1)$ be vectors in $\mathbb{R}^3$.
 - (a) (8 pts.) Find the volume of the parallelepiped with a vertex at the origin and edges X , Y and Z .
 - (b) (7 pts.) Determine whether the three given vectors form a basis for $\mathbb{R}^3$.
4. [20 pts.] Let $A = \begin{bmatrix} 1 & 0 & 2 & 5 \\ 0 & 0 & 1 & 2 \\ 1 & 0 & 1 & 3 \end{bmatrix}$.
 - (a) (5 pts.) Find the reduced row echelon form of A .
 - (b) (3 pts.) Find a basis for the **null space** of A .
 - (c) (3 pts.) Find a basis for the **row space** of A .
 - (d) (3 pts.) Find a basis for the **column space** of A that consists of columns of A .
 - (e) (3 pts.) Find the **rank** of A and the **nullity** of A .
 - (f) (3 pts.) Determine a value for k , if any, for which the vector $U = (1, k, 2)$ belongs to the **column space** of A .
5. [20 pts.] Let $A = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$.
 - (a) (5 pts.) Show that the eigenvalues of A are $\lambda_1 = \lambda_2 = 0, \lambda_3 = 2$.
 - (b) (10 pts.) Find all the eigenvectors of A .
 - (c) (5 pts.) If possible, find a nonsingular matrix P and a diagonal matrix D such that $P^{-1}AP = D$.

6. [$5 \times 4 = 20$ pts.] In each of the following questions, select the correct answer.

I. Let A be an $n \times n$ skew symmetric matrix. If $B^T + 5A - 3I_n = 5(B - A)^T$, then:

(A) $B = I_n$

(B) $B = A$

(C) $B = O_n$

(D) $B = -\frac{3}{4}I_n$

(E) None of the above.

II. If $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, then:

(A) 0 is an eigenvalue of A

(B) The system $AX = O$ has a unique solution

(C) $\text{rank}(A) = 3$

(D) A is nonsingular

(E) None of the above.

III. If $W = \{(a, b, c) \in \mathbb{R}^3 \mid 2a + b - c = 0\}$, then:

(A) $\dim(W) = 0$

(B) $\dim(W) = 1$

(C) $\dim(W) = 2$

(D) $\dim(W) = 3$

(E) None of the above.

IV. If A is a 4×4 with eigenvalues 1, 2, 3 and 4, then:

(A) $|A| = 10$

(B) $\text{nullity}(A) > 0$

(C) $\text{rank}(A) < 4$

(D) A is diagonalizable

(E) None of the above.

Solutions

1. [10 pts.] Let $X = (1, 1, 1, 1)$ and $Y = (1, 1, 0, 1)$ be two vectors in $\mathbb{R}^4$.

(a) (3 pts.) Show that X and Y are linearly independent vectors in $\mathbb{R}^4$.

Since neither of the two nonzero vectors X and Y is a multiple of the other, they are linearly independent vectors in $\mathbb{R}^4$.

(b) (7 pts.) Find a basis for $\mathbb{R}^4$ which contains X and Y .

Let $S = \{(1, 1, 1, 1), (1, 1, 0, 1), (1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), (0, 0, 0, 1)\}$. The set S spans $\mathbb{R}^4$.

Form $c_1(1, 1, 1, 1) + c_2(1, 1, 0, 1) + c_3(1, 0, 0, 0) + c_4(0, 1, 0, 0) + c_5(0, 0, 1, 0) + c_6(0, 0, 0, 1) = (0, 0, 0, 0)$.

This leads to a homogeneous system whose augmented matrix is:

$$\left[\begin{array}{cccccc|c} 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 \end{array} \right] \rightarrow \text{RREF: } \left[\begin{array}{cccccc|c} 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & -1 & 0 \end{array} \right].$$

Since the leading 1's appear in columns 1, 2, 3 and 4, we conclude that $\{(1, 1, 1, 1), (1, 1, 0, 1), (1, 0, 0, 0), (0, 1, 0, 0)\}$ is a basis for $\mathbb{R}^4$ that contains the two given vectors.

2. [15 pts.] Let $P(1, 2, 3)$, $Q(0, 3, 4)$, and $R(1, 0, 2)$ be three points in $\mathbb{R}^3$.

(a) (8 pts.) Find the area of the triangle PQR .

Let $U = \overrightarrow{PQ} = (-1, 1, 1)$ and $V = \overrightarrow{PR} = (0, -2, -1)$. Then $U \times V = (1, -1, 2)$. Therefore, the area is $A = \frac{1}{2} \|U \times V\| = \frac{1}{2} \|(1, -1, 2)\| = \frac{1}{2} \sqrt{6}$.

(b) (7 pts.) Find an equation of the plane passing through the points P , Q and R .

Using $P(1, 2, 3)$ as a point in the plane and $N = U \times V = (1, -1, 2)$ as a normal to the plane, we get: $(x - 1) - (y - 2) + 2(z - 3) = 0$.

3. [15 pts.] Let $X = (1, -2, 4)$, $Y = (3, 4, 1)$ and $Z = (-1, 1, 1)$ be vectors in $\mathbb{R}^3$

(a) (8 pts.) Find the volume of the parallelepiped with a vertex at the origin and edges X , Y and Z .

$$V = \text{abs} \begin{vmatrix} 1 & -2 & 4 \\ 3 & 4 & 1 \\ -1 & 1 & 1 \end{vmatrix} = 39.$$

(b) (7 pts.) Determine whether the three given vectors form a basis for $\mathbb{R}^3$.

Since the above determinant is nonzero, the three vectors are linearly independent vectors in $\mathbb{R}^3$ ($\dim(\mathbb{R}^3) = 3$), hence they form a basis for $\mathbb{R}^3$.

4. [20 pts.] Let $A = \begin{bmatrix} 1 & 0 & 2 & 5 \\ 0 & 0 & 1 & 2 \\ 1 & 0 & 1 & 3 \end{bmatrix}$.

(a) (5 pts.) The reduced row echelon form of A is $\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

(b) (3 pts.) A basis for the **null space** of A is $\{(0, 1, 0, 0), (-1, 0, -2, 1)\}$.

(c) (3 pts.) A basis for the **row space** of A is $\{(1, 0, 0, 1), (0, 0, 1, 2)\}$.

(d) (3 pts.) A basis for the **column space** of A is $\{(1, 0, 1), (2, 1, 1)\}$.

(e) (3 pts.) **rank**(A) = 2 and the **nullity**(A) = 2.

(f) (3 pts.) $U = (1, k, 2)$ is in the **column space** of $A \Leftrightarrow U = (1, k, 2)$ is a linear combination of the vectors $(1, 0, 1), (2, 1, 1)$.

Form $c_1(1, 0, 1) + c_2(2, 1, 1) = (1, k, 2)$.

This leads to a linear system whose augmented matrix is $\left[\begin{array}{cc|c} 1 & 2 & 1 \\ 0 & 1 & k \\ 1 & 1 & 2 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 2 & 1 \\ 0 & 1 & k \\ 0 & 0 & k+1 \end{array} \right]$. It is

clear that the system is consistent for $k = -1$.

5. [20 pts.] Let $A = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$.

(a) (5 pts.)

$f(\lambda) = \begin{vmatrix} \lambda & -1 & -1 \\ 0 & \lambda - 1 & -1 \\ 0 & -1 & \lambda - 1 \end{vmatrix} = \lambda[(\lambda - 1)^2 - 1] = \lambda(\lambda^2 - 2\lambda) = \lambda^2(\lambda - 2)$. Therefore the eigenvalues of A are $\lambda_1 = \lambda_2 = 0, \lambda_3 = 2$.

(b) (10 pts.)

For $\lambda_1 = \lambda_2 = 0$: $(0I_3 - A) = \begin{bmatrix} 0 & -1 & -1 \\ 0 & -1 & -1 \\ 0 & -1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

Every solution to $(0I_3 - A)X = 0$ is of the form $X = \begin{bmatrix} r \\ -s \\ s \end{bmatrix} = r \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, r, s \in \mathbb{R}$.

The eigenvectors of A associated with the eigenvalue 0 are of the form: $r \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$, where

r and s are not both zero.

For $\lambda_3 = 2$: $(2I_3 - A) = \begin{bmatrix} 2 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$.

Every solution to $(2I_3 - A)X = 0$ is of the form $X = t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, t \in \mathbb{R}$.

The eigenvectors of A associated with the eigenvalue 2 are of the form: $t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, where $t \neq 0$.

(c) (5 pts.) If possible, find a nonsingular matrix P and a diagonal matrix D such that $P^{-1}AP = D$.

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \text{ and } D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

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(E) None of the above.

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(E) None of the above.